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Grade: **12B**

Subject: **Information Technology (IT)**

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PAT

Scenario

Introduction

A local property auctioneering company, Southern Auctioneers (In the Western Cape) has had trouble managing its auctioneers, property listings, buyers, sellers, and auction events.

Problem

John, the owner, has asked me to make a database application. The company specializes in residential and commercial property auctions in a specific region. They need a system to manage their property listings, buyers, sellers, and auction events.

Suggested Solutions

- Allow users to search for properties based on various criteria (location, type, price range, etc.).
- Allow online registration for buyers and bidding during auctions. Perhaps use google cloud?
- Generate reports on auction results, buyer activity, and seller performance.
- Implement a user interface for data entry, retrieval, and modification for each table.
- Implement security measures to protect sensitive data. - Passwords and Usernames, Codes, or cell phone numbers.
- Allow the manager or owner to change databases.
- Auctioneers cannot delete or insert auctions, only receptionists and owners.
- Allow receptionists, owner, admins and auctioneers to filter and search the database.
- Calculate the salary/pay of each auctioneer based on their commissions and salary respectively, this varies, depending on the contract between the auctioneer and client.
- Create a graph of the auctioneer's salary.

Scope

- Generate reports on auction results, buyer activity, and seller performance.
- Implement a user interface for data entry, retrieval, and modification for each table.
- Implement security measures to protect sensitive data. - Passwords and Usernames, Codes, or cell phone numbers.
- Allow the manager or owner to change databases.
- Calculate the salary/pay of each auctioneer based on their commissions (7%).
- Create a graph of the auctioneer's salary.

- Allow receptionists, owner, admins and auctioneers to filter and search the database.
- Auctioneers cannot delete or insert auctions, only receptionists and owners.

Bibliography

- Gemini (research)
- Property24 (research and examples)
- Mockaroo (used for the database)

User Stories

Owner (John) = Own

Administrator = Admin

Receptionist = Recep

Auctioneer = Auction

 - Noun

 - Verb

 - Key Notes

Owner

User Stories			Conversations & Confirmations		
As a	I want to be able to....	So that I can....	Success	Failure	P/F
Own	Login	View each auctioneer's commission or salary. Name, surname, cell number, username, Password and Job position. Validate the username and password has no spaces. Validate that the password has the correct length. (checked)	Forms change	Message: Text file not found.	
Own	See last time logged in	Ensure my own privacy	Message: Last logged in at [Date]	Message: Text file not found.	

Own	Logout	Prevent others from using the account. Save all changes.	Forms change	Forms do not change	
Own	Exit	Stop others from trying to login to accounts.	Application terminates	Application does not terminate	

Administrator

User Stories			Conversations & Confirmations		
As a	I want to be able to....	So that I can....	Success	Failure	P/F
Admin	Login	Add/Delete users	Forms change	Text file not found.	
Admin	See last time logged in	Ensure my own privacy	Message: Last logged in at [Date]	Message: Text file not found.	
Admin	Manage users	Add new users, username and password (correct length) may not have spaces or special characters	Message: User added	Messages: Please make sure there are no spaces	
Admin	Manage users	Edit users, usernames, passwords (correct length) and cell numbers may not have spaces or special characters	Message: User's information has changed.	Messages: Please make sure there are no spaces	

Admin	Manage key parts of the program	View access log. Loaded from text file [Users.txt]	Access log is displayed.	Error reading file: Text file not found.	
Admin	Logout	Prevent others from using the account. Save all changes.	Forms change	Forms do not change	
Admin	Exit	Stop others from trying to login to accounts.	Application terminates	Application does not terminate	

Receptionist

User Stories			Conversations & Confirmations		
As a	I want to be able to....	So that I can....	Success	Failure	P/F
Recep	Manage properties and auctions	Add new events/auctions This will include the Auction: date auctioned, time auctioned, location, status Seller: ID, cell no., name, surname, email, property address Auctioneer: ID, cell no., name, surname	Successful addition	Message: Please make sure there are no spaces	

Recep	See last time logged in	Ensure my own privacy	Message: Last logged in at [Date]	Message: Text file not found.	
Recep	Manage properties and auctions	Edit properties and sellers	Successful change to event/property	Message: Please make sure there are no spaces	
Recep	Logout	Prevent others from using the account. Save all changes.	Forms change	Forms do not change	
Recep	Exit	Stop others from trying to login to accounts.	Application terminates	Application does not terminate	

Auction

User Stories			Conversations & Confirmations		
As a	I want to be able to....	So that I can....	Success	Failure	P/F
Auction	Login	View each auctioneer's commission	Forms change	Message: - Text file not found. - No sale found assigned to you	

Auction	See last time logged in	Ensure my own privacy	Message: Last logged in at [Date]	Message: Text file not found.	
Auction	Only see assorted auctions with the user as the assigned auctioneer.	Ensure other auctioneers privacy	The auctions will be displayed: Top: Completed Middle: Upcoming Bottom: Ongoing	Message: MDB file not found.	
Auction	Manage sales	Edit ongoing and upcoming auctions.	Message: Sale has been changed successfully	Message: Please make sure there are no spaces	
Auction	Choose which ongoing auction to auction	Auction my selected auction.	Message: Auction Form closes and Auction2 opens	-	
Auction	Logout	Prevent others from using the account. Save all changes.	Forms change	Forms do not change	
Admin	Exit	Stop others from trying to login to accounts.	Application terminates	Application does not terminate	

Auction2

User Stories			Conversations & Confirmations		
As a	I want to be able to....	So that I can....	Success	Failure	P/F
Auction	See last time logged in	Ensure my own privacy	Message: Last logged in at [Date]	Message: Text file not found.	
Auction	Only see auctions with the user as the assigned auctioneer	Ensure other auctioneers privacy	The auctions will be displayed: Top: Completed Middle: Upcoming Bottom: Ongoing	Message: MDB file not found.	
Auction	See reserve bid, start/current bid and property description.	Auction correctly and efficiently.	See reserve bid, start/current bid and property description.	Message: Text file not found.	
Auction	Input buyer information: name, surname, cell phone number, ID, Email and confirmation that	Save the auction	Message: Sale has been Completed	Message: Please make sure there are no spaces (in cell number)	

	the buyer accepts the T&C. Check if the buyer's information is in the correct format				
Auction	Logout	Prevent others from using the account. Save all changes.	Forms change	Forms do not change	
Auction	Exit	Stop others from trying to login to accounts.	Application terminates	Application does not terminate	

Journal Repository: Noun-Verb Analysis

Candidate :	Class :	Method :	Attribute :	Comment :
TUsers	x			Each user's account will be an object, hence being an array of objects. Each user's information (attributes) will be read from a text file named "Users.txt".
Login		x		The login will check if the username and password matches, if so, the forms will change according to the job position
LengthCheck		x		Used to validate if the password is the correct length (>10)
SpaceCheck		x		Used to validate if the username and password has no spaces
User_Name			x	An attribute to the User class, stored in "Users.txt" and used in the array of objects in the Login method.

User_Surname			x	An attribute to the User class, stored in “Users.txt” and used in the array of objects in the Login method.
User_CellNumber			x	An attribute to the User class, stored in “Users.txt” and used in the array of objects in the Login method.
Username			x	An attribute to the User class, stored in “Users.txt” and used in the array of objects in the Login method.
Password			x	An attribute to the User class, stored in “Users.txt” and used in the array of objects in the Login method.
JobPosition			x	An attribute to the User class, stored in “Users.txt” and used in the array of objects in the Login method. Will be owner, admin, receptionist or auctioneer.

SalesType			x	An attribute to the User class, stored in "Users.txt" and used in the array of objects in the Login method. This will only have a value for auctioneers
SeeLoggedIn		x		Allow any user to see their last login, read from a text file
ChangeAuctioneer Dept		x		Used to change the amount owed to each auctioneer by the owner
Logout		x		Change forms to login window
Exit		x		Terminates program
SpecialCharCheck		x		Checks of the username and password has no spaces or special characters
First		x		Go to the first record/row in the database grid

Last		x		Go to the last record/row in the database grid
Next		x		Go to the next record/row in the database grid
Previous		x		Go to the previous record/row in the database grid
Insert		x		Insert a record in the database
Edit		x		Edit a record in the database
Delete		x		Delete a record in the database
ViewAccessLog		x		The administrator can see the full access log, read from a text file and into a memo.
Date_Auctioned			x	A field in the Auctions table
Time_Auctioned			x	A field in the Auctions table

Location			x	A field in the Auctions table
Status			x	A field in the Auctions table
Seller_ID			x	A field in the Seller's table
Seller_Cell_Number			x	A field in the Seller's table
Seller_Name			x	A field in the Seller's table
Seller_Surname			x	A field in the Seller's table
Seller_Email			x	A field in the Seller's table
Property_Address			x	A field in the Properties table
Auctioneer_ID			x	A field in the Auctioneer table

Auctioneer_Cell_number			x	A field in the Auctioneer table
Auctioneer_Name			x	A field in the Auctioneer table
Auctioneer_Surname			x	A field in the Auctioneer table
Sorted_Assigned_Auctioneer		x		A procedure that sorts the auctions belonging to the user's assigned auctions. Ongoing auctions will be at the bottom followed by upcoming auctions in the middle and completed auctions at the top
Auction		x		Forms will change from the Auction form to the Auction2 form, where the details for the selected ongoing auction will be read and displayed in the Auction2 form
Reserve_Bid			x	A field in the Auction's table
Start_Bid			x	A field in the Auction's table

Bid_Increment			x	A field in the Auction's table
Current_Bid		x		A label that displayed the current bid amount
Property_Description			x	Text loaded from a text file
Buyer_Name			x	A field in the Buyer's table
Buyer_Surname			x	A field in the Buyer's table
Buyer_Cell_Number			x	A field in the Buyer's table
Buyer_ID			x	A field in the Buyer's table
Buyer_Email			x	A field in the Buyer's table
ID_Check		x		Checks if the ID is the correct length, valid DOB and overall format

Email_Check		x		Checks if the email
Cell_Number_Check		x		Checks if the cell number is in the correct format('0'00 000 0000)
Save		x		Saves all of the seller's information

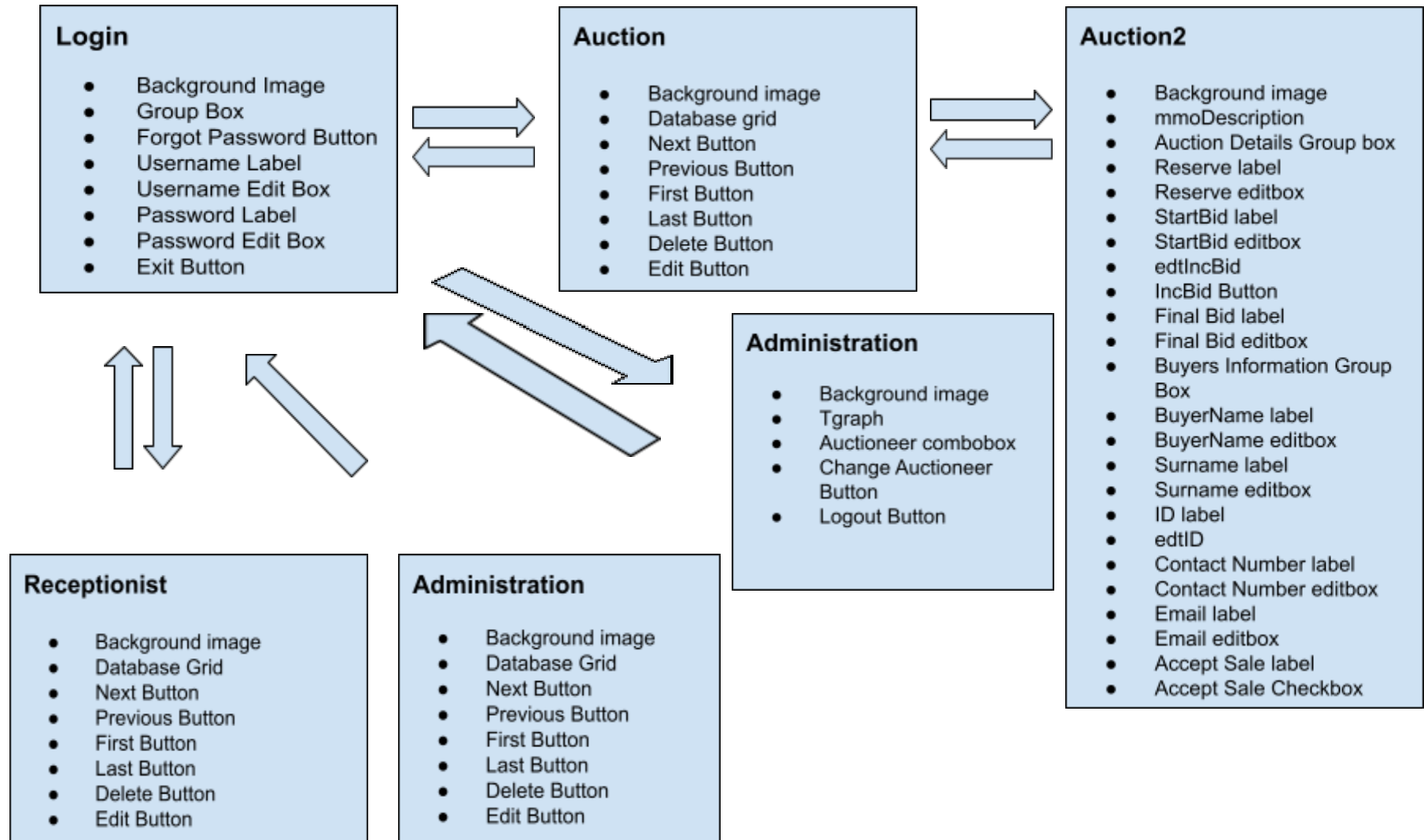
Data Dictionary

Field name	Data Type	Example	Comments (eg key)
sUsername	String	deonharm9999	Used to store the username of the user
sPassword	String	#Password3	Used to store the password of the user
tUsers	Text File [Text]	Deon#Harmse#0678986213#deonharm4634#ltPassWord4634#Auctioneer	This will store the users information in the format: Name#Surname#CellNumber#Username#Password#JobPosition
tPropertyDescription (Property ID as name)	Text File [Text]	Description	This will store all the properties descriptions. Things not in the DB, e.g gardens, area type, sea view
tUserlog ("Property" + Auction ID Field of property as name + ".txt", e.g. "Property.txt1")	Text File [Text]	Will store a user's ID + "#" + the datetime (date + space + time) the user logs out	7511061234567#06/11/2024 12:30
sList	TStringList	Deon	Used to store a list of sLine from tUsers text file
sLine	String	Deon#Harmse#0678986213#deonharm4634#ltPassWord4634#Auctioneer	Will be used to copy and deleted parts of the string based on the "#"
sTemp	String	"Deon"	Used for parts of string
iLoop	Integer	5	Used to loop through databases
bFound	Boolean	True	Used to stop searching through database

iPos	Integer	3	Used to determine the position of a delimiter in a string
iBathrooms	Integer	2	Used to store the number of Bathrooms
iBedrooms	Integer	1	Used to store the number of Bedrooms
iGarages	Integer	3	Used to store the number of Garages
dSquareMeters	Double	165,65	Used to store the square meters of the house area
sUser_Name	String	Deon	Used to store the user's name
sUser_Surname	String	Harmse	Used to store the user's surname
sReadUsername	String	deonharm9999	Used to store the read username
sReadPassword	String	#Password3	Used to store the read password
sJobPosition	String	Administrator	Used to store the user's job position
sSalesType	String	Salary	Salary or commision
dDateAuctioned	Date	01/01/2024	Used to store the auction's date
tTimeAuctioned	Time	17:00	Used to store the auction's time
sLocation	String	9 Star Street	Used to store the location
sStatus	String	Ongoing	Used to store the status of the auction
sSeller_ID	String	1201015555063	Used to store the seller's ID
sSeller_Cell_Number	String	0123456789	Used to store the seller's cell phone number
sSeller_Name	String	Deon	Used to store the seller's name
sSeller_Surnam	String	Harmse	Used to store the seller's surname
sSeller_Email	String	deonharm9999@hsbredasdorp.	Used to store the seller's email address

		co.za	
sProperty_Address	String	10 Star Street	Used to store the seller's property address
sAuctioneer_ID	String	1201015555063	Used to store the auctioneer's ID
sAuctioneer_Cell_Number	String	0123456789	Used to store the auctioneer's cell phone number
sAuctioneer_Name	String	Deon	Used to store the auctioneer's name
sAuctioneer_Surname	String	Harmse	Used to store the auctioneer's surname
cReserve_Bid	Currency	5000000	Used to store the auction's reserve bid
cStart_Bid	Currency	1000000	Used to store the auction's start bid
cBid_Increment	Currency	50000	Used to store the auction's bid increment
sProperty_Description	String	Located in an excellent area on the Western Coast...	Used to store the properties description
sBuyer_Name	String	Deon	Used to store the buyer's name
sBuyer_Surname	String	Harmse	Used to store the buyer's surname
sBuyer_Cell_Number	String	0123456789	Used to store the buyer's cell phone number
sBuyer_ID	String	1201015555063	Used to store the buyer's ID
sBuyer_Email	String	deonharm9999@hsbredasdorp.co.za	Used to store the buyer's email address

Screen Navigation



Database Design

Properties Table

Field	Data Type	Example	Key
Property_Address	String	"910 Green Crossing"	Primary Key
Property_Type	String	"Residential" or "Commercial"	
Square Footage(m ²)	Integer (rounded off)	350	
Bedrooms	Integer	2	
Bathrooms	Integer	3	
Garages	Integer	2	
Starting bid	Integer	1 000 000	
Reserve price (optional)	Currency	2 500 000	
Auction_Event_ID (foreign key linking to	Integer	5	Foreign Key [Linked to Auctions Table]
Sellers_ID	String	"8011231234567"	Foreign Key [Linked to Sellers Table]

Buyers Table

Field	Data Type	Example	Key
Buyers_ID	String	2211239876623	Primary Key
Buyers_Cell_Number	String	Cell Phone number e.g "0634789812"	
Buyers_Name	String	Donald	
Buyers_Surname	String	Trump	
Buyers_Email_Addre ss	String	DonaldTrump@gmail .com	

Sellers Table

Field	Data Type	Example	Key
Sellers_ID	String	2211239876623	Primary Key
Seller_Cell_Number	String	Cell Phone number e.g "0634789812"	
Sellers_Name	String	Will	
Sellers_Surname	String	Smith	
Sellers_Email	String	WillSmith@gmail.co m	
Property_ID	String	"910 Green Crossing"	Foreign Key [Linking to Properties Table]

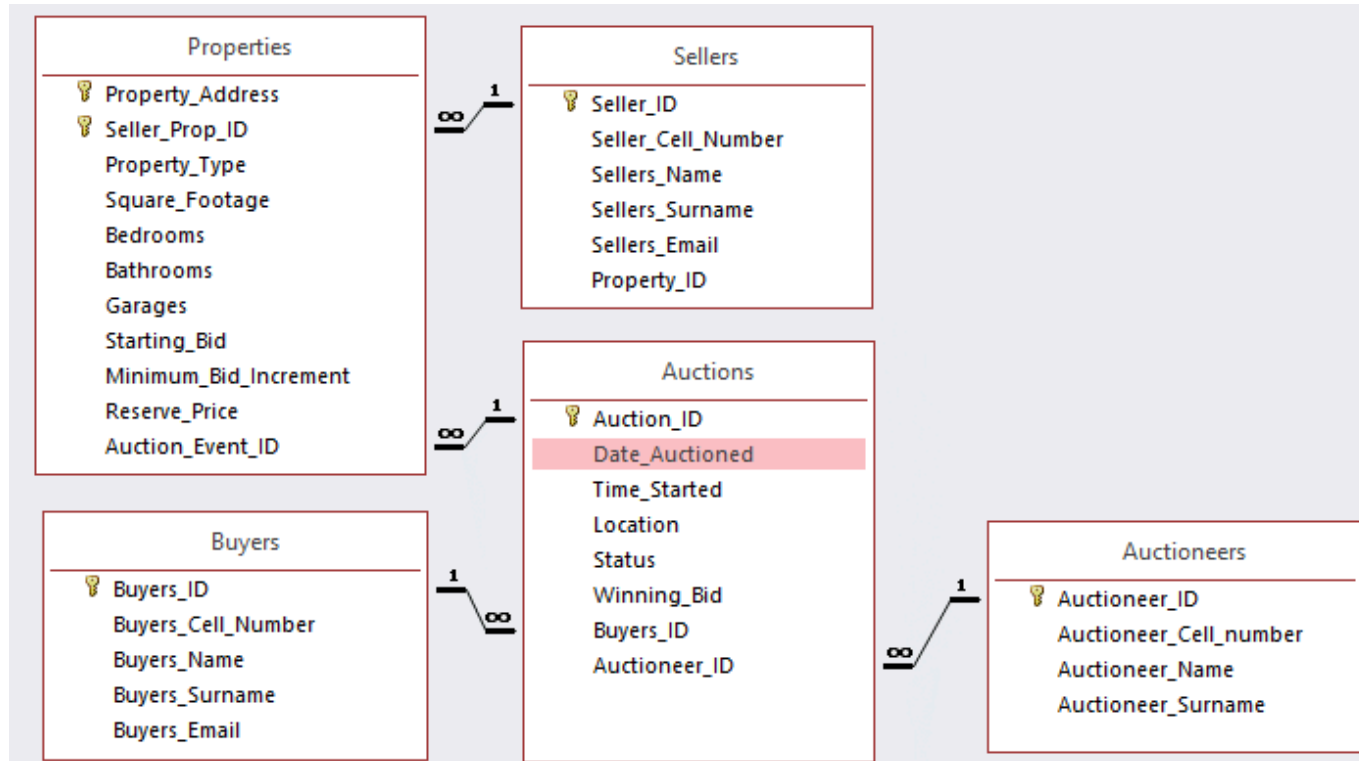
Auctions Table

Field	Data Type	Example	Key
Auction_ID	Integer	5	Primary Key
Date_Auctioned	Date	yyyy/mm/dd format e.g "2024/06/05"	
Time_Started	Time	24 hour format e.g "17:00"	
Location	String	"438 Springview Circle"	
Status	String	"Upcoming" OR "Ongoing" OR "Completed"	
Winning_Bid	Currency	R 1500 000	
Buyers_ID	String	Cell Phone number e.g "0634789812"	Foreign Key [Linking to Buyers Table]
Auctioneer_ID	String	Cell Phone number e.g "0634789812"	Foreign Key [Linking to Auctioneers Table]

Auctioneers Table

Field	Data Type	Example	Key
Auctioneer_ID	String	2211239876623	Primary Key
Auctioneer_Cell_Nu mber	String	Cell Phone number e.g "0634789812"	
Auctioneer_Name	String	Sherlock	
Auctioneer_Surname	String	Holmes	

Database Diagram



Classes

TUsers Class

-fName : String // Name of user
-fSurname : String // Surname of user
-fCellNumber : String // 10 digit cell phone number of user
-fID: String //13 digit SA ID number of user
-fUsername : String // username of user
-fPassword : String // password of user
-fJobPosition : String // job position of user [Administrator, Receptionist, Owner, Auctioneer]

+ constructor create (sName, sSurname, sCellNumber, sID, sUsername, sPassword, sJobPosition : string)
+ setName(sNewName : string)
+ setSurname(sNewSurname : string)
+ setCellNumber(sNewCellNumber : string)
+ setUsername(sUsername : string)
+ setPassword(sNewPassword : string)
+ setJobPosition(sNewJobPosition : string)

+ getName
+ getSurname
+ getCellNumber
+ getUsername
+ getPassword
+ getJobPosition
+ toString() : String

The purpose of this class is to help manage each individual user as an object. It allows the program to search, edit, delete and add new users.

Data Structures

Text files

Text file name	Format	Example
Users.txt	This will store the users information in the format: Name#Surname#CellNumber#Username#Password#JobPosition	Deon#Harmse#0678986213#deonharm4634#ItPassWord4634#Auctioneer
"Property" + Auction ID Field of property as name + ".txt", e.g. "Property.txt1" (Handle = tPropertyDescription)	This will store all the properties descriptions	Description (This will store all the properties descriptions. Things not in the DB, e.g gardens, area type, sea view)
Users_Log.txt	Will store a user's ID + "#" + the datetime (date + space + time) the user logs out	7511061234567#06/11/2024 12:30

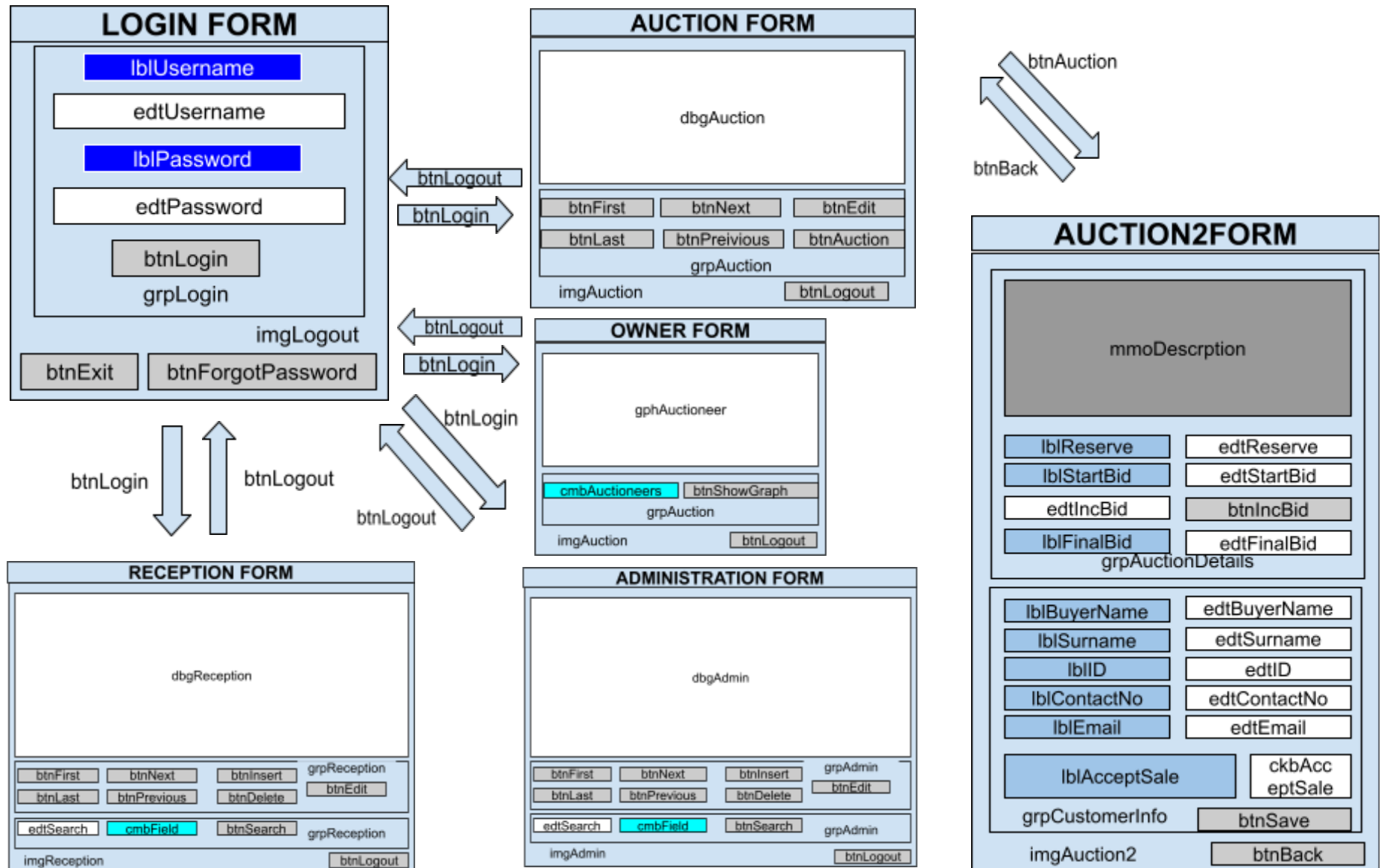
Arrays

- **Dynamic Array of Objects:** Used with each user's login. It stores the users (objects) username, password and job.
- **Dynamic Parallel Arrays:** Used to store the access log. It will store the user's name, surname, date and time when logged in.

Advanced programming

- Dynamic Arrays
- Dynamic Array of Objects
- Dynamic Parallel Arrays
- Music with timer
- Tcharts

GUI Design (Check)



IPO

Login form

Input						Processing			Output		
General			Validation (including components)			Algorithm			Layout specification (components)		
Source	Datatype	Format	What	Method	Feed Back	What	How	Feed Back	What	Format	Object
edtUsername	String	Any case	Letters/Numbers	Check username for invalid or bad chars.	'Please enter Username' / 'Please do not use special chars in Username' 'Please make sure there are no spaces'	Checks if the user is registered and current.	Locate User in a dynamic array of objects. The dynamic array of objects will be read from a text file.	On failure: 'You have not registered.' OR Focus moves to forgot password if it is the administrator or owner	'Your password/username is incorrect or you are not registered.' Focus moves to forgot password if it is the administrator or owner	Text OR Forms	Message Dialogue Box

								tor or owner			
								Success: Forms change			
edtPass word	String	Any case	Letters/Numbers	Check username for invalid or bad chars.	'Please enter Password' / 'Please do not use special chars in Password' 'Please make sure there are no spaces'	Checks if the user is registered and current.	Locate User in a dynamic array of objects. The dynamic array of objects will be read from a text file.	On failure: 'You have not registered.' OR Focus moves to forgot password if it is the administra tor or owner Success: Forms change	'Your password/ username is incorrect or you are not registered.' Focus moves to forgot password if it is the administrato r or owner that gets their password incorrect.	Text OR Forms	Message Dialogue Box

btnLogi n	On Click	-	-	-	-	Check if the username and password matches and exists	Use the locate function.	Failure: Username or password not found	Forms change	Forms	frmAucti on / frmAdmi nistration / frmRece ptionist
btnLogi n	Boolean	-	Check if a matching username and password is in the DB, use the locate function. Assign bFound a value if the matching record is found and vice-versa.	Check if bUserFou nd is true	False: Showmes sage: "Please make sure your username and password is correct"						
btnLogi n	Char	" "	Check through the username and password (string = array of chars) fo a space	Loop through the username and password and test if it is a space	True(is a space): Showmes sage: "Please make sure that you do not have any						

					spaces in you username and password”						
btnExit	On Click	-	-	-	-	-	-	-	The program is terminated	-	-
btnLogout	onClick	-	-	-	-	Changes forms, adds the user ID and datetime to userLog text file	Assign, append UserLog text file, close text file, ID stored in array of TUsers class.	Closes form, opens frmLogin	Changes to login form	Form	frmLogin frmReception

Auctions form

Input						Processing			Output		
General			Validation (including components)			Algorithm			Layout specification (components)		
Source	Datatype	Format	What	Method	Feed Back	What	How	Feed Back	What	Format	Object
on Activate	-	-	-	-	-	Only the auctioneer's auctions will be shown. Sorted by completed then upcoming and lastly ongoing.	Conditional loop Conditional loop	-	Only auctioneer's auctions are shown. Top is completed, the middle is upcoming and the bottom is ongoing.	Text	dbgAuctions
btnFirst	onClick	-	-	-	-	-	tblAuctions.first	-	The pointer moves to the first record of the grid	Text	dbgAuctions

btnLast	onClick	-	-	-	-	-	tblAuctions.last	-	The pointer moves to the last record of the grid	Text	dbgAuctions
btnNext	onClick	-	-	-	-	-	tblAuctions.next	-	The pointer moves to the next record of the grid	Text	dbgAuctions
btnPrevious	onClick	-	-	-	-	-	tblAuctions.previous	-	The pointer moves to the previous record of the grid	Text	dbgAuctions
btnEdit	onClick	-	Letters/Numbers	Check if all the edited values have the correct format	'Please make sure there are no spaces' 'Please make sure the inputted value is a cell number'	-	tblAuctions.edit	-	Values in the database is changed	Text	dbgAuctions

btnAuction	onClick	-	-	-	-	-	-	-	Changes to Auction2 form	Form	frmAuction frmAuction2
btnLogout	onClick	-	-	-	-	Changes forms, adds the user ID and datetime to userLog text file	Assign, append UserLog text file, close text file, ID stored in array of TUsers class.	Closes form, opens frmLogin	Changes to login form	Form	frmLogin frmReception

Auction2 form

Input						Processing			Output		
General			Validation (including components)			Algorithm			Layout specification (components)		
Source	DataType	Format	What	Method	Feed Back	What	How	Feed Back	What	Format	Object
on Activate	-	-	-	-	-	All the auctions details (start bid, bid increment, reserve bid and property description)	Everything read in from the database, except the property description is read from the property description text file	-	All the read information is loaded into its respective objects	Text	edtReserve edtStartBid edtBidInc

btnBidIncrement	onClick	-	-	-	-	The current bid amount is updated	The bid increment is added to the current bid. Formula: (iCurrentBid = iCurrentBid + iBid)	-	The current bid is updated on the edtCurrent bid object	Text	edtCurrentBidbtn
btnBidDecrement	Integer	-	Check that a bid decrement does not go below the reserve price	If iCurrentBid - iBid Increment < iReservePrice then show message else Decrement bid	True: Showmessage: "You cannot go below the property's reserve price"						
btnSave	onClick	-	Checks if all the information is in the correct format: Buyers: Name, Surname, ID,	SpaceCheck / SpecialCharacterCheck/ IDCheck/	'Please do not use special chars in Username' 'Please make sure	The Buyers information is added to the database	dbgAuctions.insert	-	The sale is added to the database Forms change from the Auction2 form	Form dbAuctions	Auction Auction2

			Contact Number, Email Address	CellNumberCheck Methods	there are no spaces'				to the Auction form Auction status is updated from ongoing to completed		
btnLogout	onClick					Changes forms, adds the user ID and datetime to userLog text file	Assign, append UserLog text file, close text file, ID stored in array of TUsers class.	Closes form, opens frmLogin	Changes to login form	Form	frmLogin frmAuction2

Administration form

Input						Processing			Output		
General			Validation (including components)			Algorithm			Layout specification (components)		
Source	Datatype	Format	What	Method	Feed Back	What	How	Feed Back	What	Format	Object
on Activate	-	-	-	-	-	Only the auctioneer's auctions will be shown. Sorted by completed then upcoming and lastly ongoing.	Conditional loop Conditional loop	-	Only auctioneer's auctions are shown. Top is completed, the middle is upcoming and the bottom is ongoing.	Text	dbgAuctions

btnFirst	onClick	-	-	-	-	-	dbaAuctions .first	-	The pointer moves to the first record of the grid	Text	dbaAuction s
btnLast	onClick	-	-	-	-	-	dbaAuctions .last	-	The pointer moves to the last record of the grid	Text	dbaAuction s
btnNext	onClick	-	-	-	-	-	dbaAuctions .next	-	The pointer moves to the next record of the grid	Text	dbaAuction s
btnPrevious	onClick	-	-	-	-	-	dbaAuctions .previous	-	The pointer moves to the previous record of the grid	Text	dbaAuction s

btnInsert	onClick	-	Letters/Numbers	Check if all the edited values have the correct format	'Please make sure there are no spaces' 'Please make sure the inputted value is a cell number'		dbgAuctions.Insert	-	The pointer moves to the previous record of the grid	Text	dbgAuctions
btnDelete	onClick	-	-	-	-	-	dbgAuctions.delete	-	The pointer moves to the previous record of the grid	Text	dbgAuctions
cmbField	ItemIndex	-	Check that an item (field) has been selected.	Check that the itemindex is > -1	Fail/False: Showmessage: "Please make sure you have selected a field"						

btnLogout	onClick					Changes forms, adds the user ID and datetime to userLog text file	Assign, append UserLog text file, close text file, ID stored in array of TUsers class.	Closes form, opens frmLogin	Changes to login form	Form	frmLogin frmReception
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Reception form

Input						Processing			Output		
General			Validation (including components)			Algorithm			Layout specification (components)		
Source	Datatype	Format	What	Method	Feed Back	What	How	Feed Back	What	Format	Object
on Activate	-	-	-	-	-	Only the auctioneer's auctions will be shown. Sorted by completed then upcoming and lastly ongoing.	Conditional loop Conditional loop	-	Only auctioneer's auctions are shown. Top is completed, the middle is upcoming and the bottom is ongoing.	Text	dbgAuctions

btnFirst	onClick	-	-	-	-	-	tblAuctions.first	-	The pointer moves to the first record of the grid	Text	dbgAuctions
btnLast	onClick	-	-	-	-	-	tblAuctions.last	-	The pointer moves to the last record of the grid	Text	dbgAuctions
btnNext	onClick	-	-	-	-	-	tblAuctions.next	-	The pointer moves to the next record of the grid	Text	dbgAuctions
btnPrevious	onClick	-	-	-	-	-	tblAuctions.previous	-	The pointer moves to the previous record of the grid	Text	dbgAuctions
btnEdit	onClick	Letters /Numbers	Check if all the edited values have the correct format	Loop through to find spaces, check if	'Please make sure there are no spaces'	-	tblAuctions.edit	-	Values in the database is changed	Text	dbgAuctions

				the inputted ID's date exists. Check iD and cell no. length.	'Please make sure the inputted value is a cell number'						
btnInsert	onClick	Letters /Numbers	-	Check if all the edited values have the correct format	'Please make sure there are no spaces' 'Please make sure the inputted value is a cell number'		tblAuctions.Insert	-	The pointer moves to the previous record of the grid	Text	dbgAuctions
btnDelete	onClick	-	-	-	-	-	tblAuctions.delete	Message: "Record has been deleted successfully."	The pointer moves to the previous record of the grid	Text	dbgAuctions

btnSearch	onClick	-	Checks if a field to filter/search in has been selected.	Checks if the itemindex of cmbField is >= 0 Check if the edtSearch is not empty (length)	-	Searches for the property in the database that matches the edtSearch	Use locate function to search for the value in the database	All (if any) records that match the selected field in the database, will be displayed	All records that match will be displayed	Records	dbgReception
btnLogout	onClick					Changes forms, adds the user ID and datetime to userLog text file	Assign, append UserLog text file, close text file.	Closes form, opens frmLogin	Changes to login form, login successfully recorded in text file	Form	frmLogin frmReception

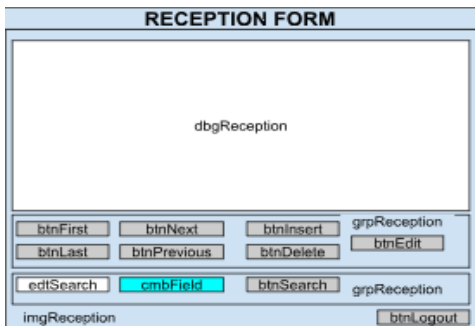
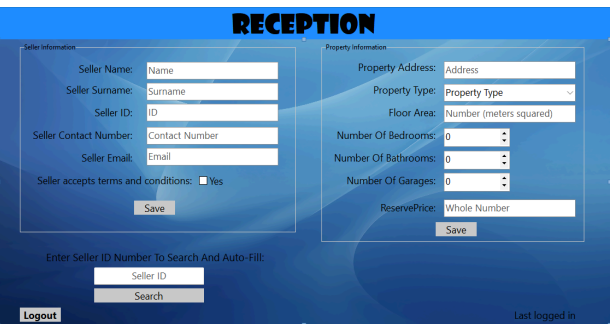
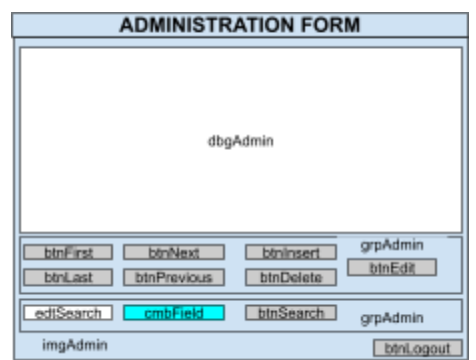
Owner Form

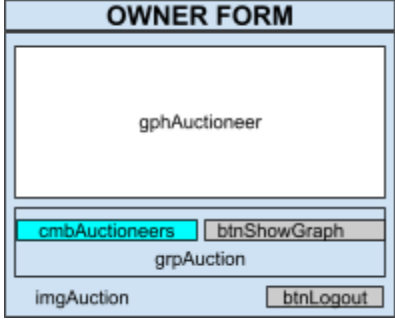
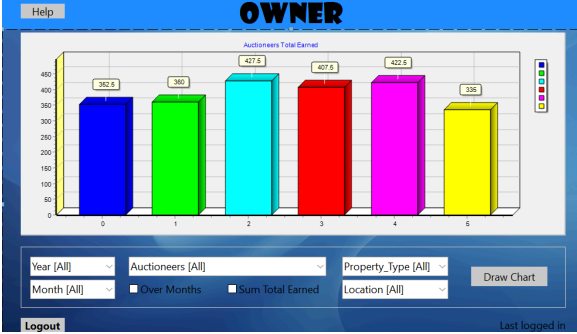
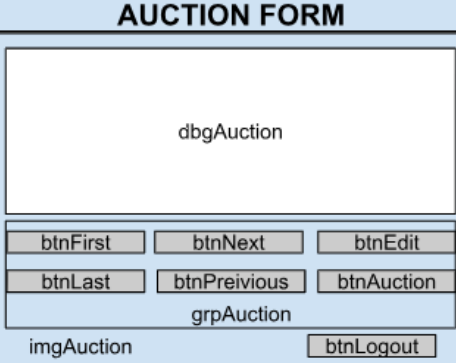
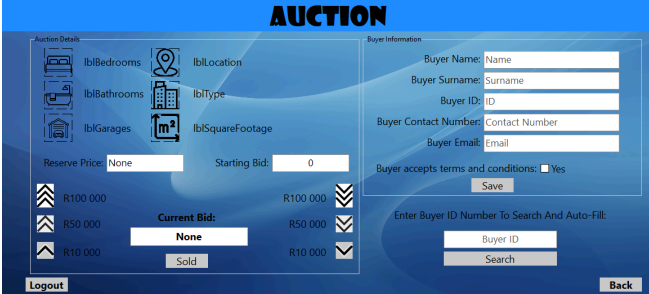
Input						Processing			Output		
General			Validation (including components)			Algorithm			Layout specification (components)		
Source	Datatype	Format	What	Method	Feed Back	What	How	Feed Back	What	Format	Object
on Activate	-	-	-	-	-	The Chart will automatically load all auctioneers and their total commission.	Use a select query for all auctioneers	False: Show message; "Auctioneers not found"	A graph is drawn, showing all auctioneers and their total commission.	Tchart	dbChart
btnDrawChart	OnClick	-	Check if the owner has chosen an auctioneer from the combo box	Check if the itemindex is ≥ 0	False: Show Message ; "Please select an auctioneer"	A dynamic query is made using the combobox input, using the chosen auctioneer ID. This is used	Look for the auctioneer matching the ID in the DB. The following	False: "Auctioneer not found in DB"	A graph is drawn, showing the chosen auctioneer and their total commission.	Tchart	dbChart

						to determine the auctioneers commision	formula is used: (0.07 * SUM([Winning_Bid])) Where the auctioneer was the auctioneer of a auction				
btnLogout	onClick					Changes forms, adds the user ID and datetime to userLog text file	Assign, append UserLog text file, close text file, ID stored in array of TUsers class.	Closes form, opens frmLogin	Changes to login form	Form	frmLogin frmReception

Change Log

- In the database, a composite key was made in the property table (Seller_ID + Auction_ID + Property_ID), as a property can be recorded several times, except with renovations, this might change other fields of the record. This might lead to changing previous records to the updated/changed property information. This would then record false information. With a composite key, multiple records of the same property can be made, allowing for the correct recording of the state of the property.
- Auctioneers can search for auctions using many variables through a query. The grid was discarded for an “auto-fill” type form when going through the database.
- When in the auction2 form (during an actual auction), the property description has been discarded as it would take too much time to implement. I have decided that rather, the basic information in the database will be displayed through labels and NOT a memo which is read through a text file.
- The Users class was changed as many attributes were not used. The only attributes used were fID, fUsername and fPassword. Each with its own getter and setter. It will all be used in the admin form, when changing and viewing each user's ID, username and password. The ID getter will be used in displaying the last login date.
- Each user's username and password was encrypted for the privacy of each user.
- The Property_ID field was removed from the Seller table as there is no use for it.
- Music with a timer was removed as the time did not accommodate it.
- Receptionist form has an ID search instead of a field filter as if a person has registered before, they should not have a problem supplying an ID for the search.
- I discovered that ID's can be changed when a person changes their ID. This would be a tedious and long task to implement for a small minority. I have decided against implementing it.

Original	New	Reason
 The original 'RECEPTION FORM' is a simple, light blue interface. It features a large central text area labeled 'dbgReception'. Below this, there are two rows of buttons: the first row contains 'btnFirst', 'btnNext', 'btnInsert', and 'grpReception'; the second row contains 'btnLast', 'btnPrevious', 'btnDelete', and 'btnEdit'. At the bottom, there is a search section with 'edtSearch', a cyan 'cmbField', 'btnSearch', 'grpReception', 'imgReception', and 'btnLogout'.	 The new 'RECEPTION' form is a more modern, blue-themed interface. It is divided into two main sections: 'Seller Information' and 'Property Information'. The 'Seller Information' section includes fields for Name, Surname, ID, Contact Number, Email, and a checkbox for 'Seller accepts terms and conditions: Yes', with a 'Save' button below. The 'Property Information' section includes fields for Address, Property Type (dropdown), Floor Area (text), Number Of Bedrooms (spinner), Number Of Bathrooms (spinner), Number Of Garages (spinner), and Reserve Price (text), with a 'Save' button below. At the bottom, there is a search section labeled 'Enter Seller ID Number To Search And Auto-Fill:' with a 'Seller ID' field, a 'Search' button, a 'Logout' button, and a 'Last logged in' status.	The new form is more intuitive and less complicated. The old is much more complex. Also allows for the linking of the seller and property table.
 The original 'ADMINISTRATION FORM' is a simple, light blue interface. It features a large central text area labeled 'dbgAdmin'. Below this, there are two rows of buttons: the first row contains 'btnFirst', 'btnNext', 'btnInsert', and 'grpAdmin'; the second row contains 'btnLast', 'btnPrevious', 'btnDelete', and 'btnEdit'. At the bottom, there is a search section with 'edtSearch', a cyan 'cmbField', 'btnSearch', 'grpAdmin', 'imgAdmin', and 'btnLogout'.	 The new administration form is a more modern, blue-themed interface. It features a top navigation bar with 'First', '<<', 'No/No', '>>', 'Last', 'Edit', and 'Insert' buttons. Below this is a search section labeled 'Enter User ID' with a 'Search' button and a 'Reset' button. The main form area includes fields for Name, Surname, User ID, Contact Number, Job Position (dropdown), Username, and Password, with 'Delete' and 'Save' buttons at the bottom. At the bottom of the form, there is a 'Logout' button and a 'Last logged in' status.	The new admin form is simpler. There are a limited number of users, hence the filter function was removed and a search according to the user ID was used instead.

Original	New	Reason
		The new form allows the owner to have more search
		The auction table has a relationship with the buyers table. This allows the adding of a buyer, which was missing in the original plan. It also allows the use of more images, a fixed auction bid change. Minimum Bid increment was therefore also removed from the properties auction table.

Demonstration Information

Username	Password	ID (Check database)	Job_Position (Form)
IdetLock4507	dettekwood6496178	4507245276496	Administrator
GaelJoci8708	GaelJocic3643432	8708275323643	Auctioneer
IngITrim8603	ebertmmell2452574	8603037492452	Auctioneer
VickSher7804	Vickyheran2877169	7804258292877	Receptionist
JillWhit7105	layneitter7558888	7105023277558	Owner

Note: Each form can logout

